

Experiment 3

Dimensional Modeling and Star Schema Design using Microsoft Power BI

Aim :

To design and implement a dimensional data model using Microsoft Power BI by transforming transactional data from the Global Superstore dataset into a Star Schema consisting of Fact and Dimension tables for efficient reporting and analytical processing.

OBJECTIVES:

- To understand the structure of a Star Schema.
- To identify suitable fact and dimension attributes from business data.
- To organize transactional data into a dimensional model.
- To create separate dimension tables for analysis.
- To connect dimension tables with a central fact table.
- To use DAX measures for analyzing business performance.
- To verify the accuracy of the developed data model.

TOOLS / SOFTWARE REQUIRED:

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor – built into Power BI Desktop
- DAX (Data Analysis Expressions)

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

Dimensional Modeling is a method of organizing data into structures that make business analysis and reporting easier. It mainly consists of **Fact Tables** and **Dimension Tables**.

Fact Table stores business transactions and measurable values such as Sales, Profit, Quantity, and Discount. It generally forms the center of the analytical model.

Dimension Tables contain descriptive information used to analyze the facts. Examples include Customer, Product, Region, and Date.

Star Schema is a dimensional model where a central fact table is directly connected to multiple dimension tables. This structure makes the data model simple, organized, and easier to use for reporting.

In this experiment, the Global Superstore Orders data is organized into a central Fact_Sales table. Separate tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date provide descriptive information for analysis.

PROCEDURE:

Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.

Launch Microsoft Power BI Desktop.

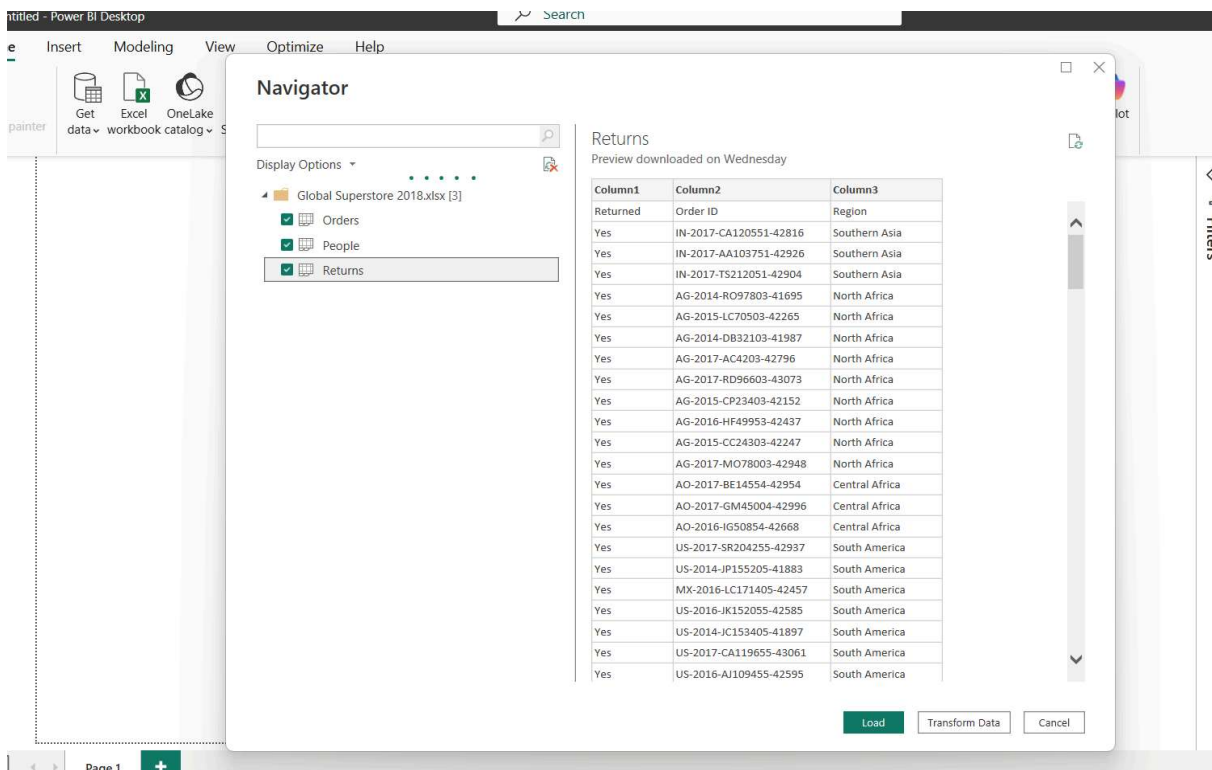
Step 2: Import the Global Superstore dataset.

Home > Get Data > Excel Workbook

Select: Global Superstore Dataset.xlsx Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:



The screenshot shows the Microsoft Power BI Desktop interface. The 'Navigator' window is open, displaying the 'Global Superstore 2018.xlsx' file. The 'Returns' worksheet is selected. The 'Returns' table preview is shown, containing the following data:

Column1	Column2	Column3
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia
Yes	IN-2017-TS212051-42904	Southern Asia
Yes	AG-2014-RO97803-41695	North Africa
Yes	AG-2015-LC70503-42265	North Africa
Yes	AG-2014-DB32103-41987	North Africa
Yes	AG-2017-AC4203-42796	North Africa
Yes	AG-2017-RD96603-43073	North Africa
Yes	AG-2015-CP23403-42152	North Africa
Yes	AG-2016-HF49953-42437	North Africa
Yes	AG-2015-CC24303-42247	North Africa
Yes	AG-2017-MO78003-42948	North Africa
Yes	AO-2017-BE14554-42954	Central Africa
Yes	AO-2017-GM45004-42996	Central Africa
Yes	AO-2016-IG50854-42668	Central Africa
Yes	US-2017-SR204255-42937	South America
Yes	US-2014-JP155205-41883	South America
Yes	MX-2016-LC171405-42457	South America
Yes	US-2016-JK152055-42585	South America
Yes	US-2014-JC153405-41897	South America
Yes	US-2017-CA119655-43061	South America
Yes	US-2016-AJ109455-42595	South America

- Orders
- Returns
- People

Verify that the imported tables appear in Power BI.

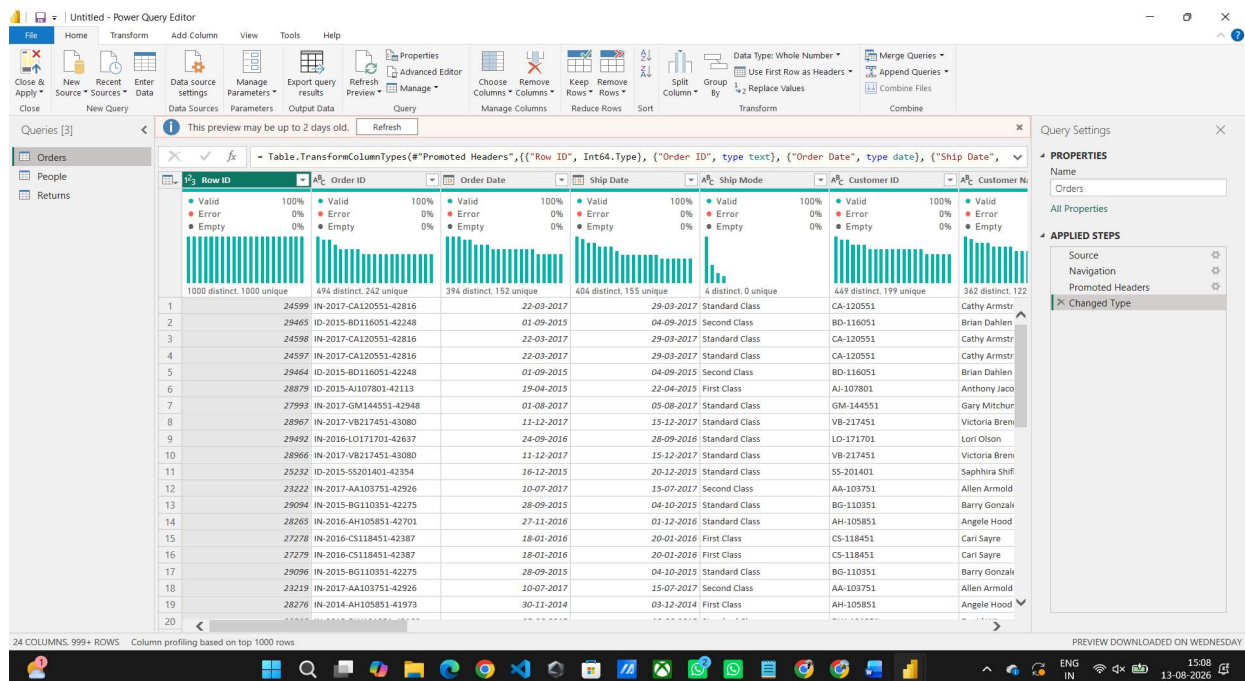
Part B: Identify Fact and Dimension Tables

Step 4: Analyse the Orders table. Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity
- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.



Identify descriptive attributes related to:

- Customer
- Product

- Region

- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data. Organize the attributes into:

A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

This classification forms the basis of the Star Schema.

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.

Home > Transform Data

Step 8: Create Dim_Customer.

The screenshot displays the Microsoft Power Query Editor. The main area shows a table with three columns: 'Customer ID', 'Customer Name', and 'Segment'. Each column has a data type of 'Text' and a 'Valid' status of 100%. The 'Customer ID' column has 1000 distinct values, 'Customer Name' has 706 distinct values, and 'Segment' has 3 distinct values. The table contains 21 rows of data. The right-hand pane shows the 'Query Settings' for 'Dim_Customer', with the 'Applied Steps' list including 'Source', 'Removed Other Columns', 'Removed Duplicates', 'Removed Blank Rows', and 'Capitalized Each Word'. The bottom status bar indicates '3 COLUMNS, 999+ ROWS' and 'Column profiling based on top 1000 rows'.

Create a customer dimension containing:

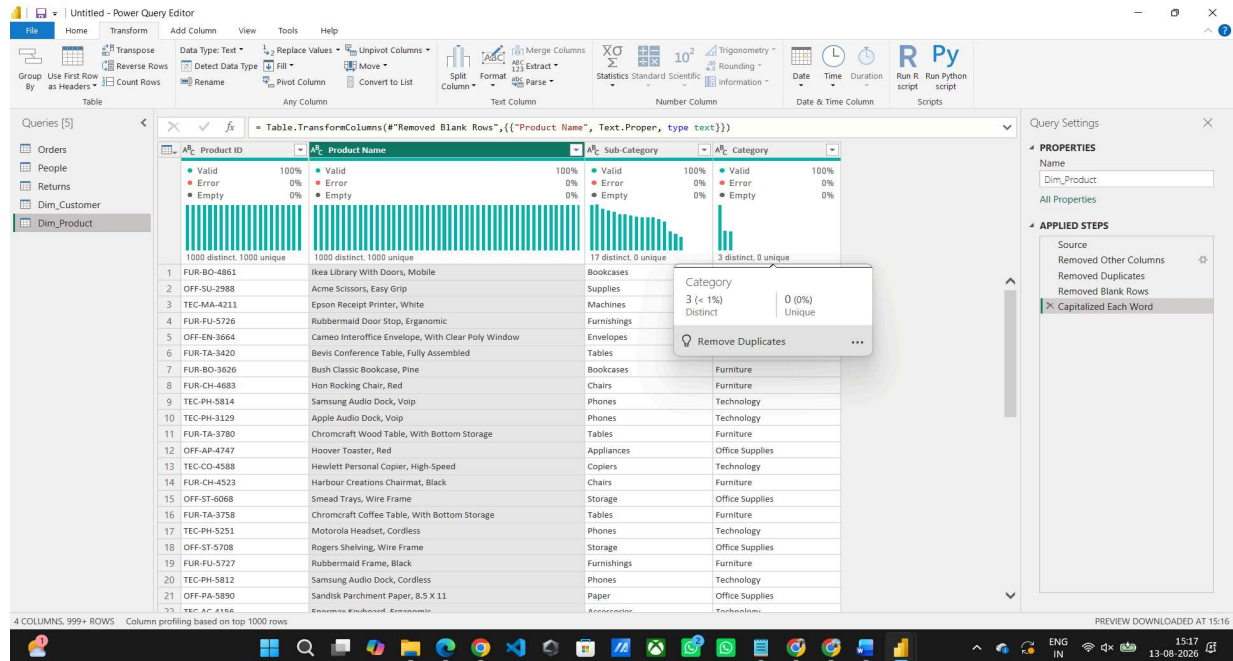
- Customer ID

- Customer Name

- Segment

Remove duplicate records from the dimension.

Step 9: Create Dim_Product.



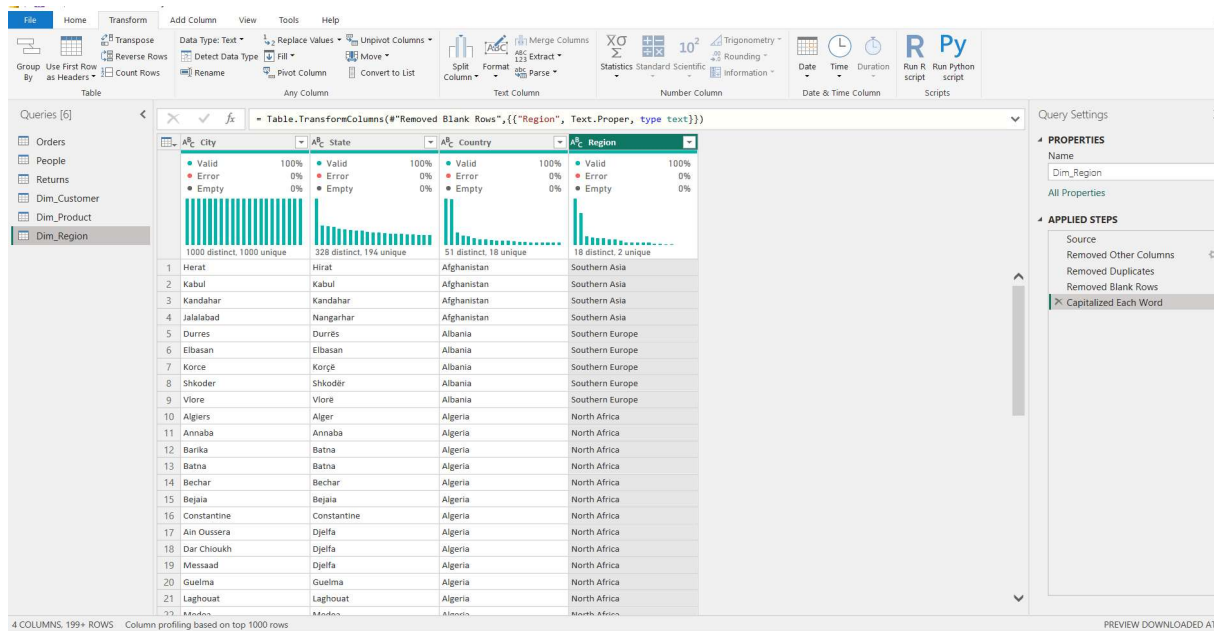
Create a product dimension containing:

- Product ID
- Product Name
- Category
- Sub-Category

Remove duplicate records.

Step 10: Create Dim_Region.

Create a region dimension containing:



- Region

- Country

- State

- City

Remove duplicate records.

Step 11: Create Dim_Date.

Create a date dimension containing:

- Order Date

- Year

- Quarter

- Month

Ensure that the date-related fields are correctly formatted.

File	Home	Transform	Add Column	View	Tools	Help
Close & Apply	New Source	Recent Sources	Enter Data	Data source settings	Manage Parameters	Export query results
Close	New Query	Data Sources	Parameters	Output Data	Query	Advanced Editor
						Choose Columns
						Remove Columns
						Keep Rows
						Remove Rows
						Sort
						Split Column
						Group By
						Use First Row as Headers
						Replace Values
						Merge Queries
						Append Queries
						Combine Files

Queries [7]	= Table.SelectRows("#Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))					
Orders	Order Date	Year	Month	Month Name	Quarter	
People	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	
Returns	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	
Dim_Customer	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	
Dim_Product	4 distinct, 4 unique	2 distinct, 0 unique	4 distinct, 4 unique	4 distinct, 4 unique	4 distinct, 4 unique	
Dim_Region	1	2017	3	March	1	
Dim_Date	2	2015	9	September	3	
	3	2015	4	April	2	
	4	2017	12	December	4	

Step 12: Remove duplicate records.

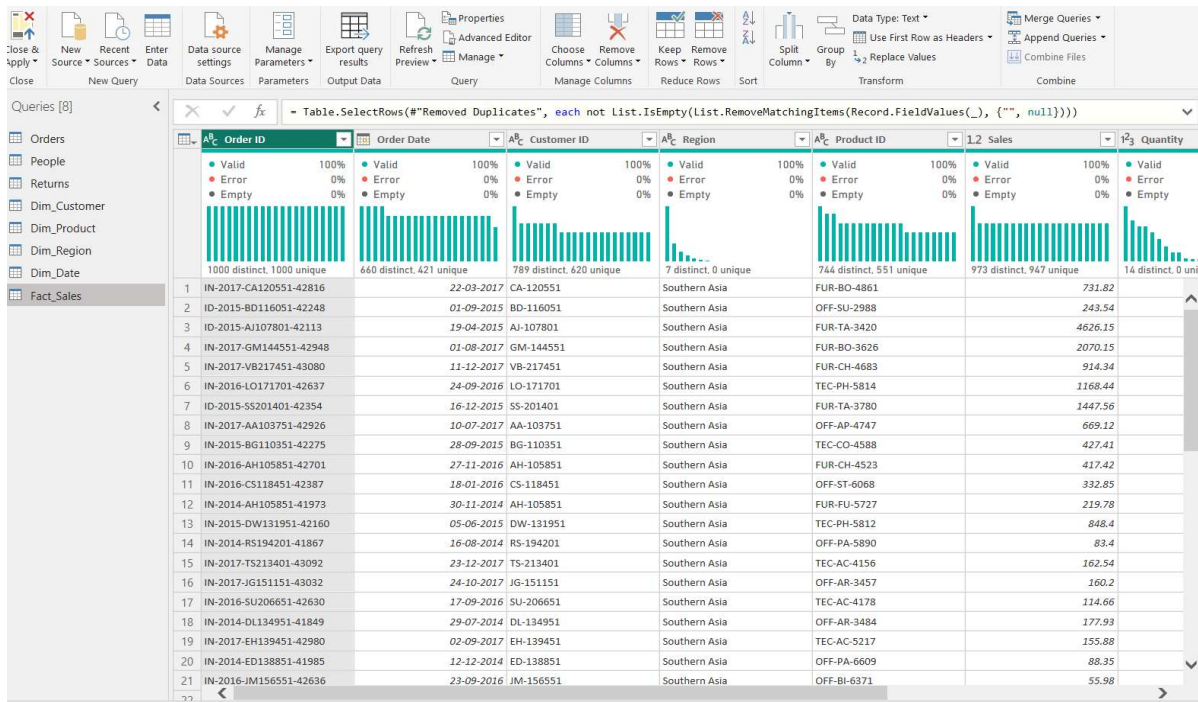
Remove duplicate records from all dimension tables and verify that the key fields contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.

Create the central Fact Table containing:

- Order ID
- Customer ID
- Product ID
- Region
- Order Date
- Sales
- Quantity
- Profit
- Discount



Step 14: Verify key columns.

Verify that all key columns required to establish relationships with the Dimension Tables are present and correctly formatted.

The resulting Fact_Sales table acts as the central table of the Star Schema.

Part E: Build Star Schema

Step 15: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.

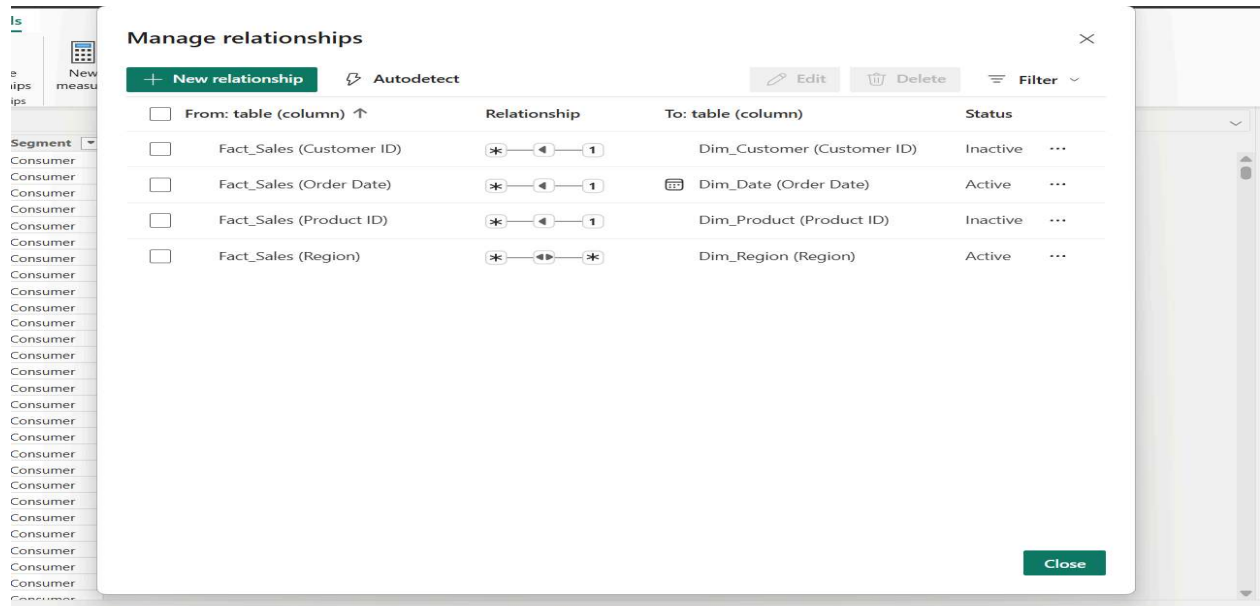
Step 16: Create relationships.

Create relationships between:

- Fact_Sales ↔ Dim_Customer
- Fact_Sales ↔ Dim_Product
- Fact_Sales ↔ Dim_Region
- Fact_Sales ↔ Dim_Date

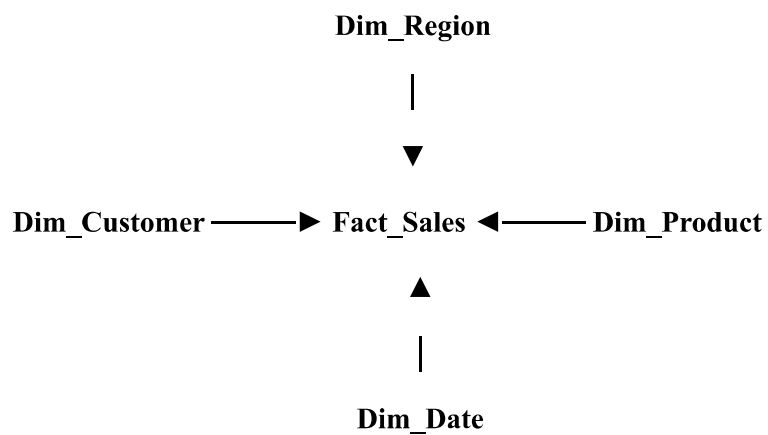
Configure relationships.

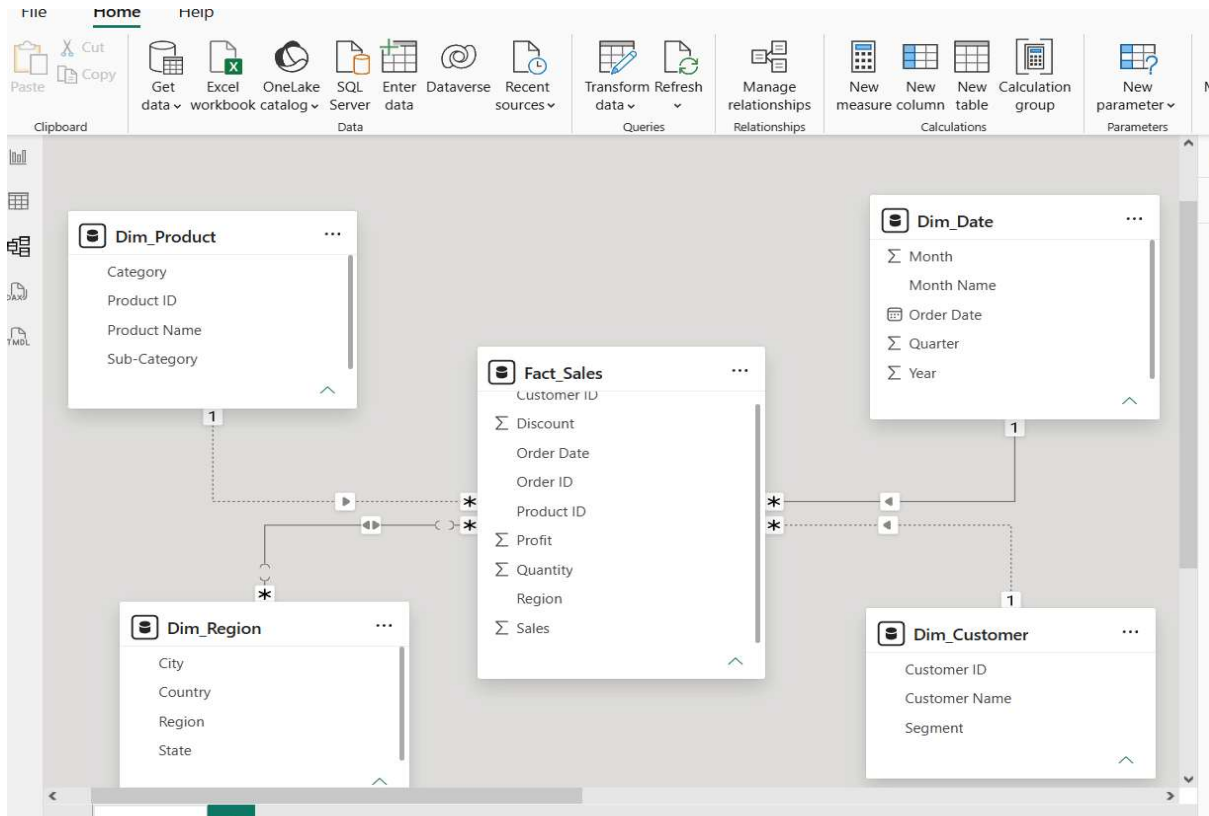
Set the relationships appropriately as Many-to-One, with the Fact Table on the many side and the corresponding Dimension Table on the one side.



Step 18: Arrange the model.

Arrange the tables visually so that:





This forms the Star Schema with Fact_Sales at the centre and the Dimension Tables . surrounding it.

Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.

Create visualizations using the dimensional model to analyse the data.

Examples include:

- Sales by Category
- Profit by Region
- Sales Trend by Year

The reference specifies these as examples of visualizations for validating the model.

Step 20: Create DAX measures.

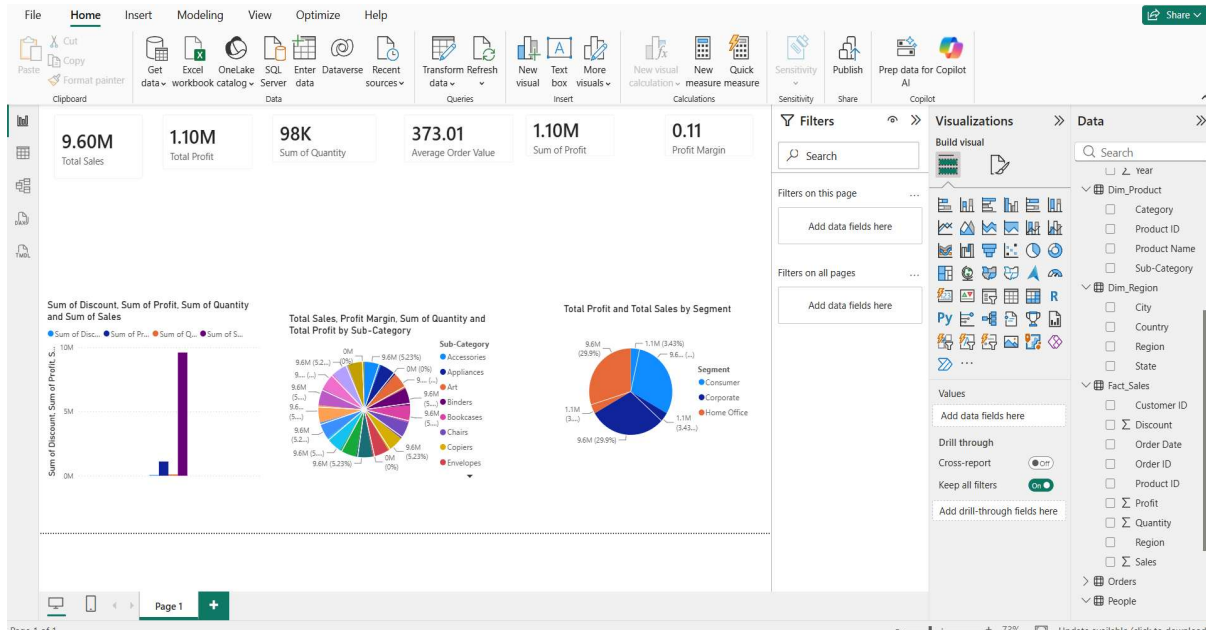
Create the following measures:

1. Total Sales = SUM(Fact_Sales[Sales])
2. Total Profit = SUM(Fact_Sales[Profit])

3. Profit Margin = $\text{DIVIDE}(\text{SUM}(\text{Fact_Sales}[\text{Profit}]), \text{SUM}(\text{Fact_Sales}[\text{Sales}]))$

4. Average Order Value = $\text{DIVIDE}([\text{Total Sales}], \text{DISTINCTCOUNT}(\text{Fact_Sales}[\text{Order ID}]))$

Step 21: Validate the Star Schema.



Use the created Dimension Tables to filter the Fact Table and verify that the relationships work correctly. Check that changes in dimensions such as Customer, Product, Region, and Date correctly affect the corresponding Fact Table data and dashboard visualizations.

RESULT:

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

LEARNING OUTCOMES:

- Understood the ETL process and its role in Business Intelligence.
- Learned how to extract data from a real-world business dataset using Power BI.
- Gained practical experience with Power Query Editor.
- Learned how to clean, filter, transform, and merge datasets.
- Learned how to handle duplicate records, blank values, and data type inconsistencies.